

These data structures allow for easy manipulation, indexing, and aggregation of data, making it suitable for various data analysis and manipulation tasks.

- **Comprehensive data analysis tools:** Pandas offers a rich set of functions and methods for data manipulation, cleaning, reshaping, and analysis. It includes functions for data alignment, missing data handling, grouping, merging, pivoting, and statistical operations, enabling users to perform complex data transformations and analysis tasks with ease.

Limitations

- Pandas' operations may be slower than low-level languages due to its focus on ease of use over raw performance.
- Pandas' operations are typically single-threaded, which can pose a significant challenge for large scale parallel processing.

Matplotlib, Seaborn, and Bokeh

Data visualisation is crucial for analysing and interpreting data, aiding in insights, pattern identification, and effective communication. In Python, several libraries offer diverse visualisation capabilities, each with unique strengths. We explore three popular libraries: Matplotlib, Seaborn, and Bokeh.

Matplotlib is widely used for creating static, interactive, and publication-quality visualisations across various domains, including scientific research and engineering.

Seaborn, built on Matplotlib, focuses on statistical data visualisation. It provides a high-level interface for creating attractive and informative statistical graphics, such as heatmaps and violin plots, commonly used in data exploration and hypothesis testing.

Bokeh is ideal for creating interactive and web-ready visualisations. It specialises in producing plots that can be embedded

into web applications and dashboards seamlessly. It is often used in web development and for building interactive data applications.

Key features of Matplotlib

- Matplotlib offers extensive customisation options for creating a wide range of plots, including line plots, scatter plots, bar charts, histograms, and more.
- It provides support for multiple output formats, including PNG, PDF, SVG, and interactive formats suitable for web applications.
- Matplotlib integrates seamlessly with Jupyter notebooks, making it ideal for exploratory data analysis and interactive data visualisation.

Key features of Seaborn

- Seaborn simplifies the process of creating complex statistical visualisations with minimal code, thanks to its high-level interface and built-in themes.
- It offers specialised functions for visualising relationships between variables, such as scatter plots with regression lines, pair plots, and joint plots.
- Seaborn provides advanced statistical functionalities, including support for grouping, aggregation, and categorical data visualisation.

Key features of Bokeh

- Bokeh excels in creating interactive and web-ready visualisations, with support for tooltips, zooming, panning, and brushing.
- It offers powerful layout options for arranging multiple plots and widgets on a single web page, facilitating the creation of interactive dashboards and data applications.
- Bokeh provides seamless integration with popular web frameworks such as Flask and Django, enabling developers to build dynamic and interactive web applications with ease.

Comparative study: When to use what Matplotlib is well-suited for creating static, publication-quality plots and visualisations, making it ideal for generating figures for scientific papers, reports, and presentations.

Seaborn is preferred for statistical data visualisation tasks, especially when exploring relationships between variables, visualising distributions, and conducting exploratory data analysis.

Bokeh is the go-to choice for building interactive web-based visualisations and data applications where user interaction and dynamic updates are required.

Limitations

- **Matplotlib:** While Matplotlib is highly customisable, its syntax and API can be verbose, leading to code verbosity and decreased productivity in some cases.
- **Seaborn:** Seaborn's specialisation in statistical visualisation limits its scope for creating non-statistical plots and visualisations, which may require using Matplotlib directly.
- **Bokeh:** Bokeh's learning curve can be steep for beginners, especially when building complex interactive visualisations and web applications. Additionally, its performance may fall when handling large datasets or rendering complex plots with a high number of data points.

scikit-learn: Streamlined machine learning for data scientists

scikit-learn is arguably the most popular open source machine learning library used by Python developers. It makes training and evaluating models easier and provides a large selection of algorithms. NumPy, SciPy, Matplotlib, and other scientific Python libraries are the foundation upon which scikit-learn is constructed. scikit-learn includes a variety of unsupervised and supervised learning techniques. Many effective techniques